

Synchronisation in data networks

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Synchronisation is of fundamental importance to the performance of digital networks, but the effects of poor network timing on the information being transported are often largely ignored. This paper provides an introduction to the subject of synchronisation, discusses the current BT network synchronisation arrangements and describes the platform being deployed to address both the current network requirements and the needs of future telecommunications technologies.

1. Introduction

Synchronisation is the term applied to the process by which the transmission rates of elements within a digital network are locked to a common frequency source. Synchronisation is generally required for digital networks in order to prevent loss and corruption of data. For network operators, the main requirement for synchronisation has historically been to limit the size of buffers required in the public switched telephone network (PSTN), but technologies such as the new synchronous digital hierarchy (SDH) transport networks also require synchronisation. All of these require switches to be synchronised to allow the transfer of data to the appropriate quality. Both PSTN and private circuit networks may also deliver timing information to customer networks.

This paper discusses the evolution of existing network synchronisation arrangements and describes a solution being deployed to address both the current network requirements and the needs of technologies such as ATM.

2. The need for synchronisation

The information content of digital data is carried through both the amplitude and phase of the transmitted signal. One great advantage of digital transmission is that it allows the original information to be easily recovered from a signal degraded by the effects of transmission. Degradation of the signal amplitude may be introduced by noise or transmission medium attenuation and the signal typically must be regenerated on a link-by-link basis. Other effects degrade the phase information of a signal as illustrated in Fig 1.

The temporal position of a pulse edge is seen to vary from the ideal position in a cyclic manner. This variation can be caused by electrical noise, optical fibre dispersion or other short-term effects in which case the variation is at a

high frequency and is known as 'jitter'. However, low-frequency effects such as diurnal changes in the effective length of a transmission path caused by temperature cycles are also present which introduce 'wander'. The two effects are similar and are distinguished only by the frequency at which they occur, with the classification changing at 10 Hz. In any given network, both effects will be present and transmission performance is degraded by a 'jitter spectrum'. Jitter and wander are usually measured in unit intervals, either peak-to-peak or r.m.s., where one unit interval is equal to the bit period of the signal being examined. The effects of high frequency jitter can be reduced through filtering, but practical filters are not ideal and so the effects tend to accumulate across a network.

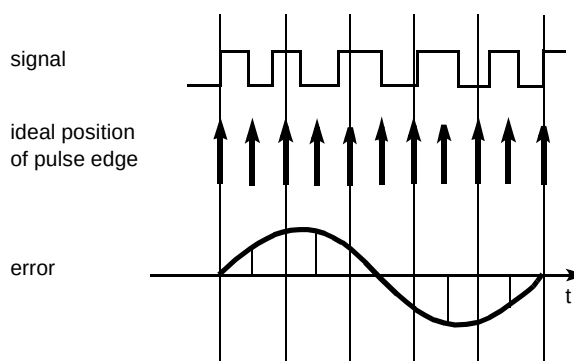


Fig 1 Digital signal timing error.

Wander, being a very low frequency effect, is difficult to measure and is usually characterised by the maximum time interval error (MTIE) or the time variance (TVAR). The former parameter is determined by sampling the time error between a measured signal and an ideal reference. MTIE is the peak-to-peak variation of these samples over the measurement period. Strictly, MTIE relates to an ideal reference (universal co-ordinated time — UTC), whereas, in

practice, a high-quality reference oscillator is used and the measurement becomes maximum relative time interval error (MRTIE). The measurements of time error may alternatively be filtered using a defined algorithm to generate the time variation (TVAR), or time deviation (TDEV) which is the square root of TVAR. The choice depends upon the application. TDEV is useful for characterising clock phase noise but is influenced by systematic effects such as diurnal wander. In such cases, e.g. in dimensioning buffers, MTIE is used.

Digital equipment typically uses an internal oscillator to generate transmitted data, the quality of which varies greatly between equipment types. The temporal quality of the digital signal, whether in terms of absolute frequency or phase stability, is of importance because any differences between interconnected digital signals must be accommodated in order to avoid slips in the data. Variations are usually absorbed by buffers, but these are expensive and lead to additional delays in the data path. The size of the buffer employed is determined by the maximum time variation which must be accommodated.

The principle is illustrated in Fig 2. Data is transmitted by a network element A under the control of an internal oscillator. The timing content of the data is extracted at receiver B and the resultant clock signal is used to write the data into a buffer store (W in Fig 2). The data is transmitted to the subsequent network element under the control of the equipment clock in node B (R in Fig 2). The buffer store in node B absorbs any degradation introduced by the transmission process and accommodates any differences in clock frequency between the two network elements.

Where several diverse paths terminate on a network element, the buffer stores have to be larger in order to accommodate greater variations in bit rate and degradation. In order to limit the size of these buffers, which are expensive and introduce delay into the transmission path, there is a need to limit the maximum difference in frequency between any two paths in a digital network. This may be accomplished by phase locking the oscillator within the equipment to a central reference clock although a residual phased-locked loop error will remain.

Synchronisation can therefore be defined as the twin requirements to ensure that:

- the cumulative effects of timing degradations introduced by the various transmission links and network elements within a network are low enough to meet the needs of the service,
- the timing frequency reference to all network elements lies within defined limits, in order to ensure that the equipment operates to its agreed specification.

For these reasons, the International Telecommunications Union (ITU) have agreed recommendations for both equipment performance, network performance, and a network reference model.

3. Equipment timing performance

Digital equipment will only tolerate a certain level of jitter on an input port before errors are introduced in the regeneration process. Furthermore, because the clock-filtering processes are not perfect, any wander present on the input will be transferred to the output. Finally, the equipment clock, being non-ideal, will also generate jitter and wander and hence these effects tend to accumulate through a chain of network elements. The effects must be limited in order to allow correct operation of the final element in the chain.

ITU Recommendation G.823 [1] defines the methodology of using jitter tolerance, jitter transfer and intrinsic jitter generation to characterise equipment timing performance although it leaves specific masks and transfer bandwidths to individual equipment recommendations. A typical jitter transfer specification is illustrated in Fig 3. In general, network equipment should attenuate jitter ($-y$) but wander is extremely difficult to filter. The equipment is therefore allowed a small jitter gain (x) of typically 0.1 dB at very low frequencies.

The tolerance of a given equipment type to jitter on its input signals is also defined internationally. Figure 4 illustrates a typical jitter tolerance specification which, for

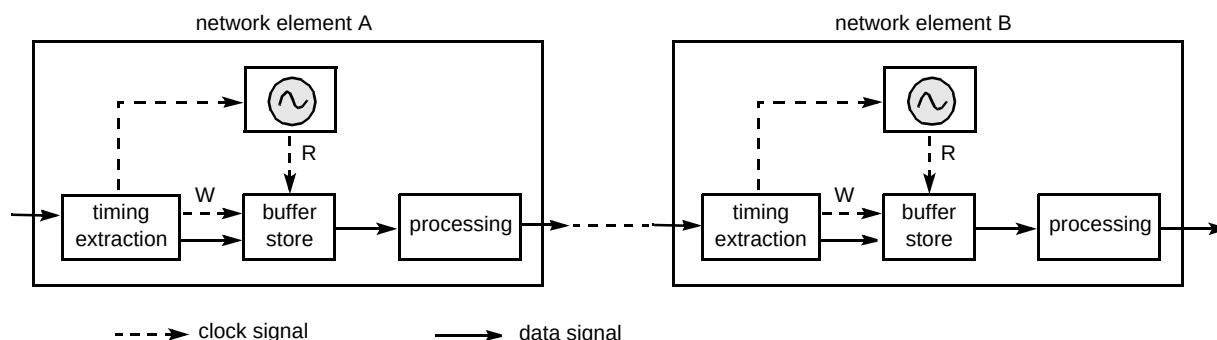


Fig 2 Link timing processes.

example, requires that an equipment should tolerate a level of jitter A_0 occurring at a frequency f_0 .

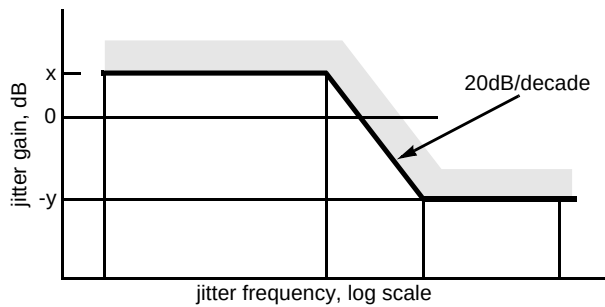


Fig 3 Typical equipment jitter transfer characteristic.

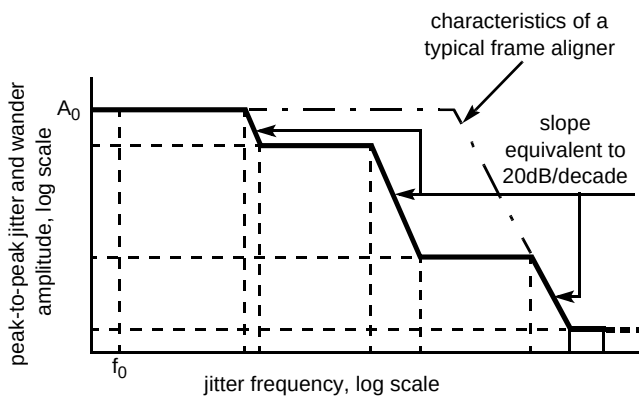


Fig 4 Typical jitter and wander tolerance mask

The performance of specific equipment is affected by the quality of timing to that network element, but the performance of a path or circuit across a network of such equipment is also affected.

In both Fig 3 and Fig 4 the frequency points (f) and the limits are dependent upon the bit rate of the signal being measured.

4. Network performance

The timing performance of paths across a network is determined by the timing performance of the network equipment. In most cases, this is defined by the 'jitter tolerance' and 'jitter transfer' characteristics of the node described above. The quality of timing within a network may be determined by the quality of any oscillators within the network measured by absolute frequency accuracy, phase noise, or frequency stability.

However, the timing performance of a complete network must also be specified; this may be through overall performance objectives on any service being transported, as in the definition of error performance and slip objectives for 64 kbit/s, or by specific timing requirements such as the control of jitter and wander at network interfaces.

Synchronisation performance is specified by the three standards bodies — ITU-T, ANSI and ETSI. Consequently this leads to different requirements on clock specifications. The architecture chosen to distribute timing to network elements determines the performance limits at the extremities of the network, the derivation of which is illustrated in Fig 5. ITU Recommendation G.823 defines the maximum jitter and wander permissible at network node data interfaces conforming to the 2 Mbit/s PDH hierarchy, which in turn defines the maximum jitter tolerance at these interfaces. However, these requirements do not apply to paths carrying synchronisation information. The derivation of the network wander model used to partition the 18 μ s wander hysteresis limit for PSTN data buffers is also given in this document. Similarly, ITU Recommendation G.825 [2] defines the jitter and wander tolerance limits for SDH data interfaces.

It should be noted that the absolute wander of a primary reference clock (PRC), as illustrated in Fig 5, has been tightened in the latest draft version of G.811 and so the model will change accordingly.

The wander reference model is used to determine the maximum levels of jitter tolerance and assumes there is a maximum relative wander at the network nodes of 18 μ s; the MTIE of the primary reference clock is 3 μ s, as in Recommendation G.811 [3]; the MTIE at the output of a network node is 10 μ s as defined in Recommendation G.824 [4]; the maximum relative wander in slave clocks is 1 μ s as in Recommendation G.812 [5].

For signals carrying a synchronisation trail more stringent limits must be applied. For example, a 2 Mbit/s interface conforming to the full jitter/wander tolerance mask in G.823 could not be used as the synchronisation source to an SDH equipment clock, if the STM-N jitter and wander tolerance limit on the output interfaces is to be met. This is because the filtering action of the SDH equipment clock cannot sufficiently suppress the jitter and wander on the 2 Mbit/s synchronisation interface. This has already been recognised within ETSI and part 3 of DE/TM-3017 [6] defines the limits on synchronisation distribution interfaces feeding synchronisation to synchronisation supply units (SSUs) and SDH equipment clocks (SECs).

5. Timing distribution

The quality of timing signal delivered to a network is determined by two factors, the quality of the primary clock within the network, and the way in which the timing is distributed from that reference.

The frequency and phase stability performance of primary reference clocks (e.g. caesium beam oscillator technology) is defined in ITU-T Recommendation G.811.

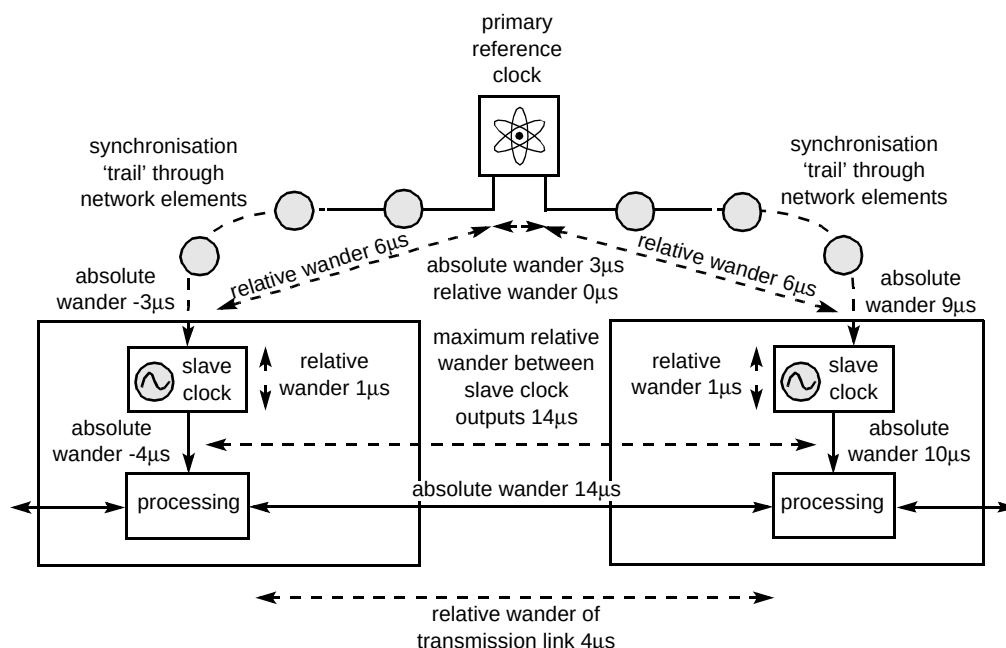


Fig 5 Wander reference model.

Improvements in technology have meant that the short-term stability requirements contained in G.811 can comfortably be achieved. The global positioning system (GPS) and other off-air standards also offer the performance of a primary reference clock. Primary reference clocks are discussed in more detail in the next section.

The architecture of synchronisation networks is agreed in ETSI and ITU documents. Two synchronisation architectures are recognised — 'master /slave' and 'mutual' synchronisation. A typical master/slave configuration is illustrated in Fig 6.

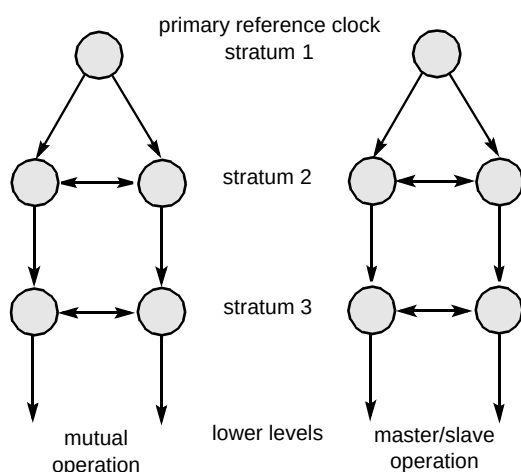


Fig 6 Digital signal timing distribution.

In this case, timing is cascaded hierarchically from a master reference clock down through the network. The quality of synchronisation is determined by the characteristics of the clocks at each level.

In mutually synchronised networks, the clocks at each level of the hierarchy are tied to provide an averaging effect as illustrated in Fig 6. Although master/slave synchronisation networks are more common, both methodologies are widely used and many network operators use a combination of the two techniques. In either case, timing information is traceable through the network from a primary reference clock via a synchronisation 'chain' which includes the physical transmission between nodes. The quality of the timing signal is degraded by transmission and improved by filtering at each layer of the distribution hierarchy. The methods for constructing synchronisation distribution chains, and the number and types of clocks that can be cascaded are standardised internationally. The route for a single synchronisation trail is illustrated in Fig 7.

A synchronisation supply unit is a timing element which contains a high-quality oscillator which may be locked to one of a number of reference inputs in order to provide an output clock exhibiting both good short-term noise performance and long-term frequency accuracy.

6. Timing references

Since the design of the original BT synchronisation network, technology improvements have been made to the various frequency references that can be used. Many of these technology improvements have been driven by the military market. This has not only influenced the types of oscillators that determine the frequency standards, but also led to the introduction of various types of time and frequency off-air receivers for use in navigation systems.

Migration of digital networks from the PDH hierarchy to the SDH hierarchy has necessitated an increase in

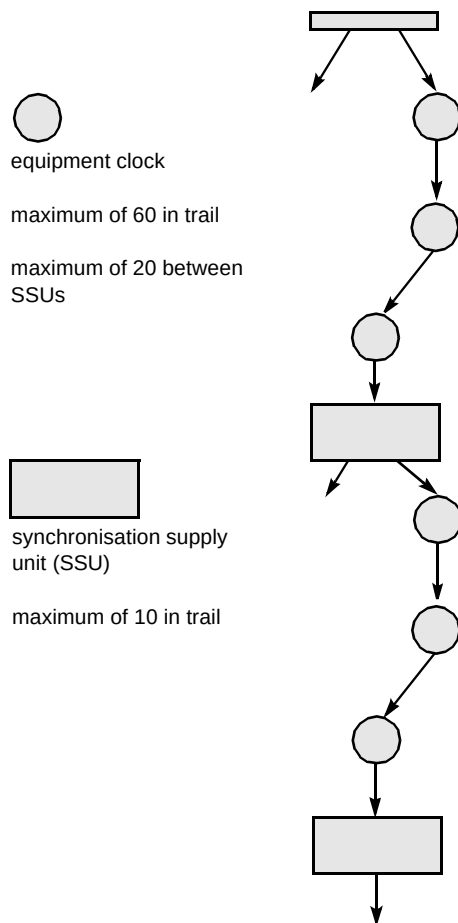


Fig 7 Synchronisation distribution trail.

performance from the synchronisation infrastructure. These requirements have provided the opportunity for specialised synchronisation manufacturers to build on the changes in technology, thus providing greater choice in the options open to provide network synchronisation.

6.1 Caesium

Caesium (Cs) beam technology is still considered to be the main frequency standard for PRC implementation. Caesium beams allow the PRC to operate independently of any other frequency source. Caesium beam technology provides a long-term frequency stability meeting the requirements of ITU Recommendation G.811. Recent advances in technology have extended the life of these devices, now allowing 8 to 10 year maintenance-free operation.

Caesium beam oscillators operate on the principle of heating atoms of Caesium 133 within a resonant tube. The movement of these atoms through the tube to a detector is controlled by magnetic fields. This magnetic resonance is then detected and used to frequency lock a crystal oscillator, thus producing a highly stable frequency source. However, during this operation the caesium atoms are used only once

and then discarded due to the operating mechanism of the tube; the tube therefore wears out and periodically needs replacing. Caesium beam tubes have the advantage that they do not frequency age, thus obviating the need for re-tuning. However, the magnetic field that controls the caesium within the tube and the crystal oscillator do require periodic adjustment on most caesium beam oscillators. Typical long-term frequency stability of the Cs beam tube is 3×10^{-12} (for the life of the Cs beam tube) and the overall system frequency is of the order of 7×10^{-12} or better, usually taking 30 minutes to reach this level of stability from cold. Cold caesium oscillators need to be switched on once a month to maintain their health. A typical life span is 5 years (dependent on caesium beam tube), although some manufacturers now produce tubes that last 10 years.

6.2 Rubidium and crystal oscillators

The performance of rubidium and high-quality crystal oscillator technology has increased with respect to cost in recent years. Combined (a mixing function) with off-air standards these provide suitable alternatives for non-autonomous PRCs.

Rubidium oscillators are often called secondary reference standards and use isotopes of rubidium 87. The atoms of rubidium are heated and resonate, this is detected and controls a crystal oscillator. As in the case of rubidium beam oscillators the isotopes of rubidium are reused, the beam does not wear out. Frequency ageing of the rubidium beam takes place and the crystal oscillator requires occasional adjustment. Typical long-term frequency stability is between 1×10^{-11} and 3×10^{-11} (100 sec observation interval) and the overall system frequency stability is of the order of 2×10^{-11} /month.

6.3 Off-air references

There are several off-air or broadcast timing signals available. This sub-section outlines the major ones.

- LORAN-C

LORAN-C is a terrestrial-based radio navigation and timing service originally owned by the US coastguard with limited world coverage. The responsibility and ownership for the transmitters has been passed to the country in which each transmitter is resident. LORAN-C may now be phased out in favour of GPS.

- OMEGA

As with LORAN-C this is a terrestrial-based radio navigation and timing service which may be discontinued in favour of GPS.

- National Physics Laboratory (NPL)

The NPL (run by the DTI) is the UK holder of time and frequency standards which can be traced back to UTC. This information is disseminated around the UK by the BT 60 kHz radio transmitter at Rugby. The typical range of these transmissions is 600 km.

- Global positioning satellites

The GPS (global positioning system) is a world wide network of satellites owned by the United States Department of Defense, primarily providing navigational and timing information for the military. A degraded source, although still very accurate, is available for commercial use and is, in fact, used by AT&T as the reference source (via rubidium beam oscillators) for their network. This service provides traceability to UTC. Each of the 24 satellites contains an ensemble of atomic clocks that are maintained by transmissions from a series of ground stations. The orbits of the satellites are such that virtually any spot on the ground is covered by at least four satellites (three to obtain position and one to transfer time). If the exact position on the ground is known then only one satellite is required to maintain time. Typical military phase accuracy to UTC (supplied by the US Naval Observatory (USNO)) is 100 ns.

The original design concept of GPS was for each satellite to transmit a clear acquisition (C/A) signal/code and a more accurate navigation signal containing the protected code (P). The C/A signal was discovered to be more accurate than intended and so the latter generation of satellites have been equipped to degrade the C/A code by selective availability (SA) for an available commercial accuracy of 250 ns to UTC-USNO. The applied SA degradation and its effect on any system which is totally reliant on GPS would need to be quantified and risks assessed. The risk that the C/A signal would be switched off in times of conflict is remote due to the level of integration in the US and world infrastructure and the loss of vital assets not navigation-related required in time of war.

- Inmarsat Geostationary Overlay (IGO)

This is a proposed timing and navigation system from geostationary satellites that will broadcast a pseudo-GPS signal with redundant coverage over most of the world. This is to be called Inmarsat Geostationary Overlay and will be used to enhance the performance of GPS for civilian navigation and timing. This service would have an accuracy approaching GPS, would be synchronised to UTC and would remove the effect of selective availability that can be currently applied to GPS.

7. BT network synchronisation

The requirements of a synchronisation network, and hence the architecture to be adopted, have changed over the years as networks have migrated to serve the needs of many services rather than a single dominant service. This section describes the evolution of the synchronisation arrangements for the BT network.

Synchronisation must be managed as a network as with any other service. The synchronisation network must be designed to deliver the quality of signal required for correct network operation, to be resilient, and to avoid timing loops in the event of failure.

The existing PSTN is synchronised via a three-level synchronisation hierarchy as shown in Fig 8. The synchronisation links are specific traffic-carrying, 2-Mbit/s PCM circuits which are currently transported over plesiochronous transmission equipment.

The levels in the existing BT synchronisation hierarchy are as follows.

- Level 1

Level 1 comprises the national reference clock (NRC) which is a triplicated caesium beam reference clock with a stability of ± 1 part in 10^{11} in accordance with the requirements of ITU Recommendation G.811. The BT NRC is currently sited in a principal digital main switching unit (DMSU) in London, termed the 'National Reference Exchange' (NRE). This unit provides the timing reference for the entire digital network and is connected by synchronisation links to a maximum of 18 selected nodes at level 2 in the hierarchy. A stand-by national reference clock is sited in a level 2 DMSU.

- Level 2

Level 2 of the hierarchy consists of all the other DMSUs in the network, currently around 65 in total, but growing due to increased core network capacity requirements. All the DMSUs in the BT network, including the level 1 node, are GPT System X digital exchanges and employ the network synchronisation subsystem (NSS) as their internal timing utility. These exchanges form a partially meshed mutually synchronised layer which can be split into a 2a and 2b level — more details are given later in this section. Each DMSU node receives between 4 and 18 synchronisation links from neighbouring nodes and up to 18 of the nodes are selected to receive links directly from the level 1 node.

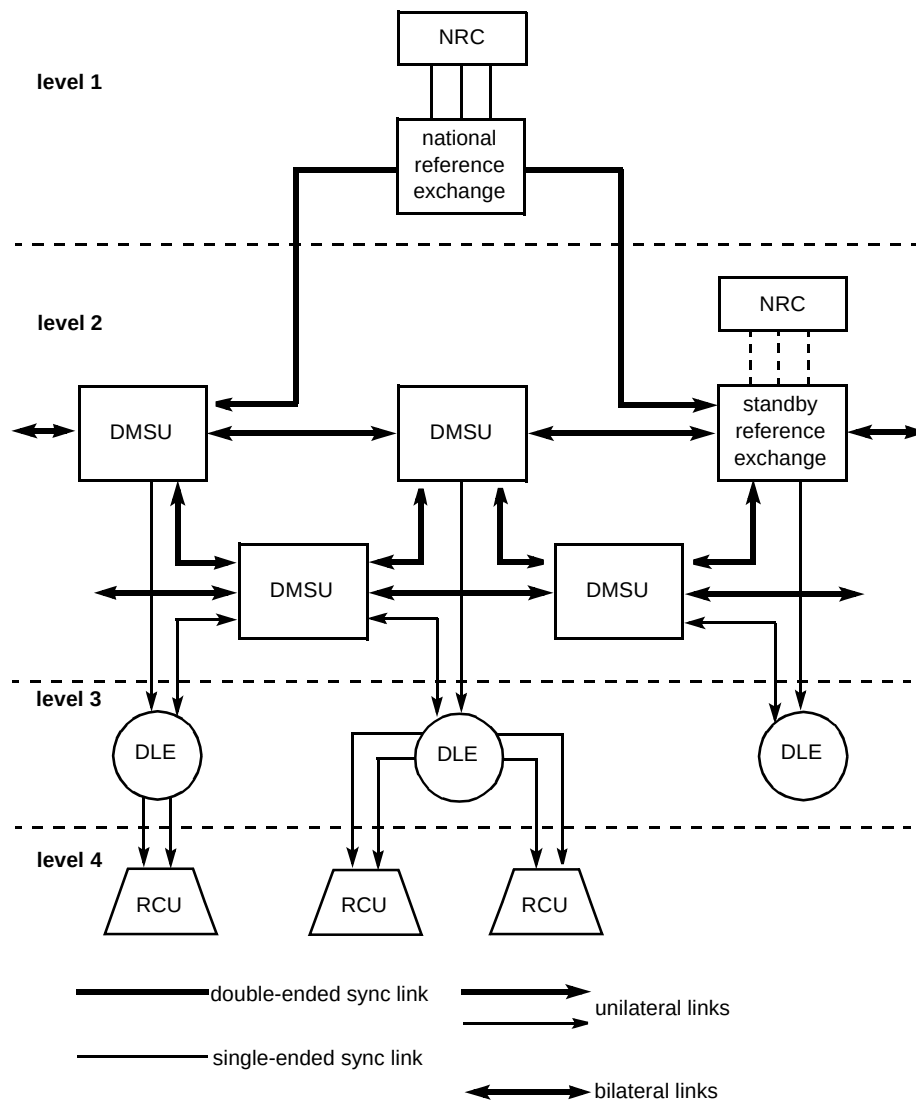


Fig 8 Original BT PSTN synchronisation hierarchy.

• Level 3

Level 3 of the synchronisation hierarchy comprises the remainder of the switched network, the digital local exchanges (DLEs) and their respective remote concentrator units (RCUs) at level 4, which number approximately 400 and 3000 respectively. The DLEs can be System X (non-NSS) or AXE10 (i.e. System Y). There are a number of AT&T 5ESS switches which are only used for the digital derived services network (DDSN). Each system has an internal synchronisation utility which is effectively slaved via a synchronisation link to its parent level 2 DMSU node. There are standby links from the parent or neighbouring DMSU provided for security.

The RCUs are similarly synchronised, slaved to their parent DLE. Northern Telecom DMS100 switches may also be employed in the network in the role of advanced service units (ASUs) for Featurenet.

Synchronisation control between level 1 and level 2 is unilateral, i.e. it is only effective at the lower level. The nodes unilaterally linked are termed 2a nodes. The links are double-ended meaning that the clock frequency control information at a node is derived not only from the local phase difference of the clock to the incoming reference signal but also from the phase difference at the remote node. Control within level 2 is bilateral between nodes and thus is effective at both ends. This enables the mutually synchronised technique to eliminate wander due to temperature effects on synchronisation links and to provide a degree of stability if both NRCs are lost. The remaining level 2 nodes with no unilateral link are termed 2b nodes. Control from level 2 to level 3 is unilateral and single-ended, i.e. master/slave; control is derived and effective at the lower node only.

This hierarchy of PSTN synchronisation is used to time all of the other networks operated by BT, including the SDH core transmission network and the entire

international backhaul network. BT also operates several major data service networks, each of which is managed independently but shares the underlying transport media. All of these networks currently take timing from 2-Mbit/s non-traffic-carrying interfaces on the NRE.

- Digital private circuit network (DPCN)

The synchronous DPCN provides end-to-end capacity (up to 2.048 Mbit/s) between customer sites. The 'permanent' circuits are routed using automatic crossconnect equipment (ACE) switches, multiplexers and equipment network access (ENA) consolidate-and-groom circuits in the access layer of the network. The DPCN synchronisation follows a traditional master/slave. The 'SyncStream' synchronisation service is provided via this network to the customer.

- Flexible bandwidth service (FBS) network

The FBS network of TDM-based core switches and multiplexers also provides end-to-end capacity (up to 2.048 Mbit/s) between customer sites. The FBS synchronisation network is split into northern and southern regions, each having a separate primary synchronisation injection point but both traceable back to the NRE. Timing is distributed in master/slave mode within regions.

- Switched multi-megabit data service (SMDS)

An asynchronous service, the SMDS network does not require synchronisation. However, where SMDS interfaces with the synchronous DPCN network to deliver the 448 kbit/s service, the SMDS switch port takes timing from the DPCN PDH interface. Additionally, as SDH transport is introduced between the core SMDS nodes, the SMDS trunks take timing from the SDH STM-N interface.

- Asynchronous transfer mode (BT CellStream service)

ATM networks can be used to deliver timing-sensitive services such as constant bit rate (CBR) and video which require accurate synchronisation. The ATM service is transported over the SDH network and is currently timed by SDH network SSUs.

- SyncStream (BT synchronisation service)

BT sells timing information to customers through a product known as *SyncStream*. This uses 64-kbit/s delivery and 2-Mbit/s presentation and offers long-term performance traceable to G.811. Although the number of customers is small, they tend to be those who operate very large private networks and who are thus of great importance to BT. The product is derived from the BT DPCN ACE switches.

Synchronisation is transported from the NRC to the customer interface in a master/slave configuration. Network synchronisation is protected down to the serving level by stand-by links which are automatically selected on failure of the priority timing feed.

The main problems with this synchronisation distribution architecture are that it is too closely bound to the PSTN technology and is therefore difficult to upgrade to meet new requirements, and the latest generation of monitoring facilities are not available. Problems with synchronisation tend to result in a degradation in performance rather than in a hard failure and are therefore difficult to both identify and isolate. The network synchronisation arrangements described above are therefore currently being replaced by a separate, managed platform to deliver synchronisation to every element in the BT digital network.

The technology now exists to allow BT to deploy a dedicated, managed transmission platform offering improvements to the quality of clock performance throughout the network. This synchronisation platform provides a secure source of synchronisation at network nodes. The platform distributes timing to the various network elements within a network node and synchronisation distribution is managed and monitored via the synchronisation platform.

The principle of the new platform is to distribute high-quality timing information throughout the network using multiple injection points rather than the single injection point of the old structure. This reduces the degradation in signal quality introduced by transmission over long systems. The platform is based upon the SSU which is designed to distribute timing information within a network node. The SSU incorporates high-quality oscillators which are locked to external reference sources to provide improved long-term frequency stability. The output of the SSU is used to provide timing to other network equipment within the node — DMSU, ACE switch, FBS switch, SDH transmission equipment, ATM switch, etc. The quality of the delivered timing signal is monitored by the SSU and the information passed to a separate management system. The external reference inputs to the SSU are a mixture of terrestrial feeds from a network primary reference clock and local references extracted from a global positioning system receiver. At least two feeds are required for each SSU in order to provide resilience and for measurement purposes.

This synchronisation equipment is currently being deployed at every major node in the BT digital network — every DMSU site and every SDH tier one site (70+ in total).

In order to provide confidence in the resilience of the new synchronisation platform, it was important to retain access to a primary reference clock. The PRC provides a

secure synchronisation source to the network. Two geographically separate nodes are being deployed to ensure security of supply for the synchronisation reference. These have been installed in the geographic centre of the UK network at Birmingham and Leicester to meet the performance standards required. The two sites are connected by both transmission and control links which allow them to logically become one reference source, although physically diverse. Comparisons of the performance of each site can also be carried out via these links. The architecture of the new platform is illustrated in Fig 9.

There are a number of options to provide the reference source for the PRC using the various frequency standards. Caesium beam frequency standards could be employed as in the current system. However, it is no longer necessary to provide three caesium beam frequency sources at each site

for security. The current generation of caesium frequency standards offers a much longer life span and so it has been possible to reduce the number of caesium standards at each site. This reduces both procurement and life-time maintenance costs. This implementation allows the BT PRC to be an autonomous source of network synchronisation and operate independently of other reference sources.

The number of caesium beams in the overall PRC configuration has been reduced with other reference standards taking their place. An off-air receiver is used at each site which is combined with duplicated oscillators to obtain the long-term performance required for international standards compliance. It would be possible to make the synchronisation platform wholly dependent on off-air standards which would also provide a traceable UTC reference source.

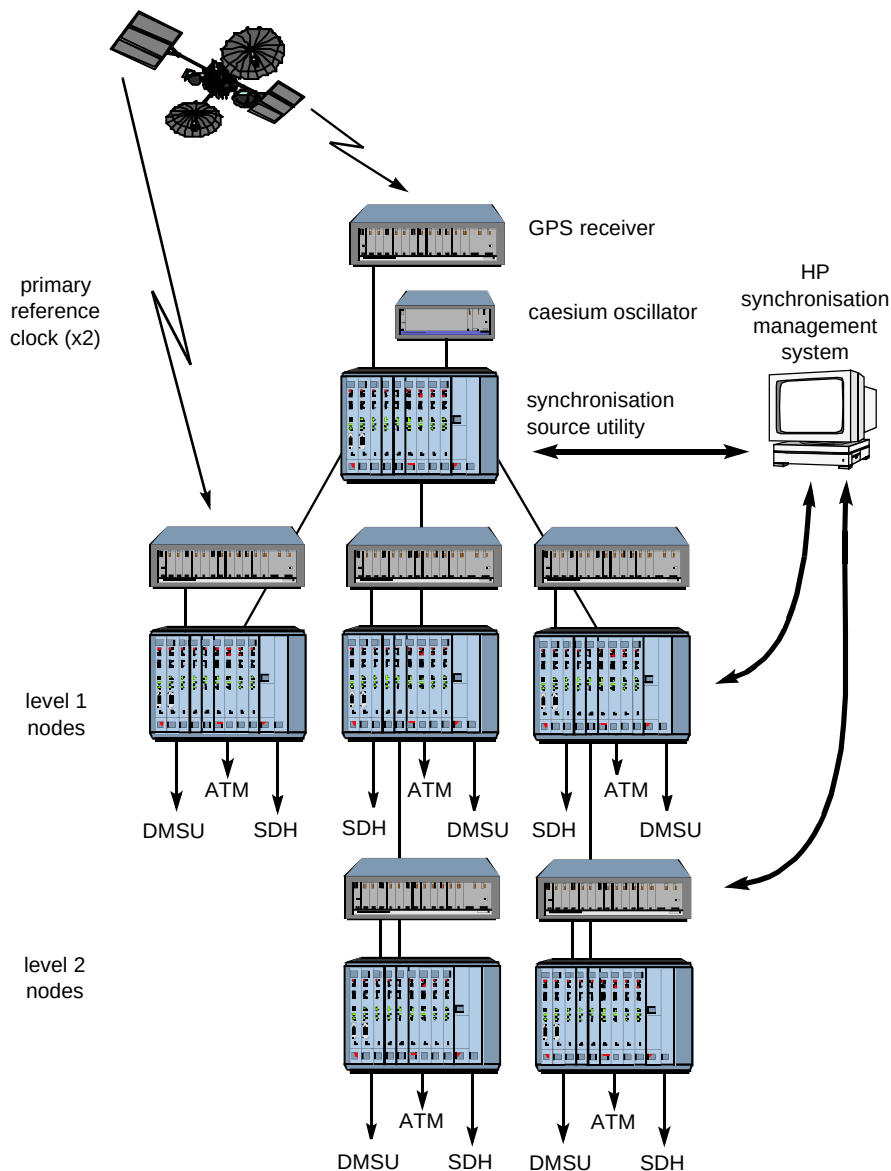


Fig 9 New synchronisation platform architecture.

The platform sites other than the PRCs are designated level 1 and level 2 in the new synchronisation hierarchy.

Complete failure of off-air references such as GPS is unlikely. Although individual receivers may be 'jammed', increased commercial use of the GPS system by airline companies and telecommunications operators makes military interference with the system less likely. However, in the event of any such failure the network has been designed to revert to the hierarchical type of operation currently used.

Synchronisation distribution to the lower level platforms in the synchronisation hierarchy would be via the transport layer, typically via the SDH bearer network. The quality of timing signal in the lower levels of the hierarchy will be further enhanced by a programme to upgrade the oscillators within all of the System X DMSUs.

Off-air synchronisation distribution has advantages in that the same quality of synchronisation can be received at all nodes. This allows all network nodes with suitable receivers, oscillators and control software to operate with PRC quality.

Distribution via space-based off-air timing, such as GPS, potentially gives the same synchronisation performance on a world-wide basis.

8. SDH network synchronisation

SDH networks are being deployed to form the transport infrastructure for BT's future network. The effects of poor synchronisation performance within the SDH transport layer on timing-sensitive services may be profound.

Figure 10 illustrates the actions of an SDH transport network in the event of a loss of synchronisation traceability. The node at which synchronisation is lost continues to receive data at the network rate. However, data is transmitted at the node rate which will, over time, differ from the network rate. The difference in data rate is by pointer movements. These are simple counters which indicate the relative position of the data from the SDH overhead. These can be either incremented or decremented to accommodate negative or positive frequency variations. This movement allows fixed padding data to be inserted to accommodate an increased node transmission rate or extra client data to be transmitted to accommodate a decreased node transmission rate. Pointers exist at two levels, the SDH administrative unit and tributary unit (TU), and the level at which pointer movements will be generated depends upon the level of processing within the network element.

For a fixed offset between the node and network clock rates, pointers will be generated at a fixed rate as illustrated in Fig 10.

These pointer movements introduce a phase movement into the transported data. In theory, these pointers would be

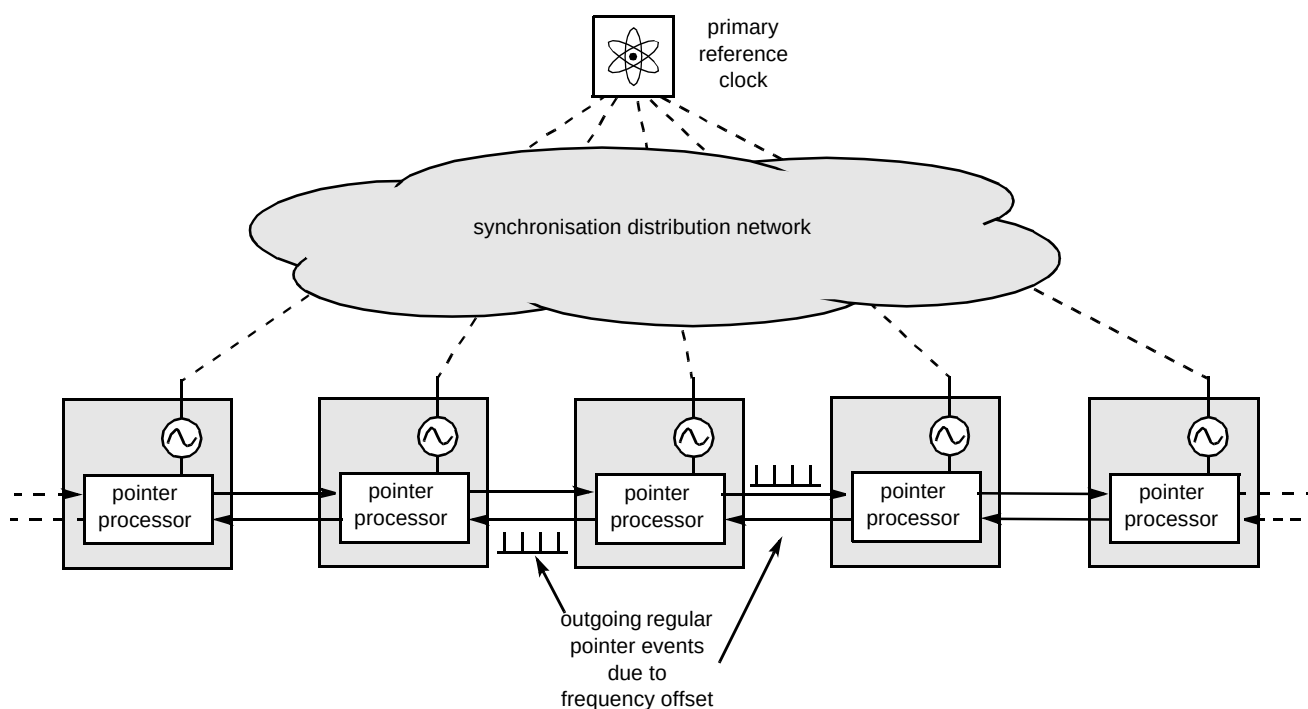


Fig 10 SDH pointer generation through synchronisation failure.

absorbed in buffers at the next network element which is locked to the network clock. However, in practice, especially where the path terminates at the next network element, the phase movement is passed on to the client service.

The effect of a single tributary unit pointer on the client data is manifest by the introduction of a phase variation. Figure 11 illustrates the phase movement introduced into 2048 kbit/s data by a single TU pointer movement.

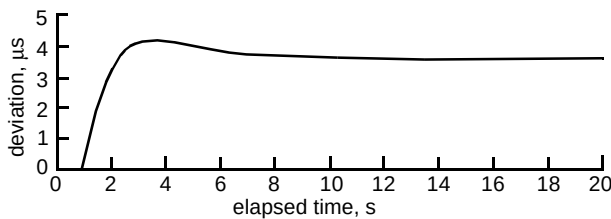


Fig 11 TU pointer-induced phase variation.

The effect of this phase movement depends upon the service being carried. The phase movement does not result in any bit errors and so the effect on speech traffic is negligible. Similarly, if the path is being used to carry synchronisation information for a 'low-quality network', typically between PABXs, then the movement is easily accommodated by the phase locked loop within the timing extraction circuitry.

However, if the path is being used to carry timing information between switches requiring high-quality timing information, the phase movement may cause the switch to reject the path as a suitable timing source.

It is therefore important that SDH transport networks are correctly synchronised and BT has applied significant design skills to ensure excellent synchronisation arrangements for the BT SDH network.

9. Synchronisation effects on other data services

Network synchronisation is required essentially in order to limit the size of buffers required for timing-sensitive services. As the quality of network synchronisation is degraded, those services which are timing sensitive are the first to be adversely affected — initially by increases in the levels of jitter and wander introduced by the transport network and ultimately by losses of data.

In general, the effects of worsening network timing performance on a service are determined by the application. Many data services, for example, are quite tolerant of poor network synchronisation. Network synchronisation consequences only adversely affect timing-sensitive services, such as voice and video, as the network component of the timing performance becomes a significant

part of the total. However, the buffers required to absorb the effects of network poor synchronisation introduce delay into the transport networks and this can adversely affect even timing-tolerant services.

As described above, the timing performance of paths across a digital network is determined by the performance of the network equipment. This effect comprises two components — the frequency difference between the transmitting and receiving equipment, and the variations introduced by the transmission network.

9.1 Voice

Within a communications network, voice switches are generally locked to a central primary reference clock and hence the differences in frequency are limited between source and destination of an inter-switch transmission link. Buffers are installed in voice switches to absorb any wander or jitter introduced by the transmission link, and any residual frequency differences between different clocks. However, because these buffers introduce delay to which voice services are intolerant, these buffers are of a finite size.

The effect of poor synchronisation on voice circuits is that these buffers are filled, and controlled frame slips are introduced which are deliberate losses of data. These slips are perceptible to the user as 'clicks' and the slip buffers are dimensioned to meet service objectives agreed for inter-operator links (where each operator's network is synchronised to a separate PRC). In order to ensure that voice switches are locked to a central reference oscillator, timing may be carried between switches using PDH transmission systems. The pointer mechanism used to accommodate differences in clock frequencies across SDH networks introduces phase discontinuities which prevent the use of such traffic links for the transportation of timing information without the introduction of additional buffers. For SDH networks, the optical bearer signal provides an enhanced platform for the transport of timing signals.

9.2 Data

Routers and other data CPE may typically be interconnected by point-to-point links to form networks. Where such links are transported over PDH equipment which can tolerate a data frequency variation of ± 50 parts per million at 2048 kbit/s, the CPE may operate in a master/slave or a dual-master mode where each device imposes timing on its transmitted data. PDH links form a timing-transparent connection between such devices, but SDH networks can introduce phase variations as previously described. The effect of these phase movements on most data CPE is negligible as the oscillators within the CPE are not of sufficient quality to track the movement.

9.3 Video

Video services require good synchronisation between end nodes but large buffers absorb many of the effects of poor network synchronisation. However, the phase discontinuities introduced by SDH pointer movements can severely disrupt video services as they disrupt the rate of change in phase of the colour sub-carrier. Again, these problems can be overcome by using additional buffers to absorb the phase movement.

10. Conclusions

Good-quality synchronisation of digital networks is vital to improve the quality of such networks.

Improving the quality of synchronisation in a digital network improves the performance of the services being transported by that network.

BT is currently deploying a new, dedicated platform to deliver improved timing performance to every element in the network.

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